

Commission Decision Management System

PROJECT OVERVIEW

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1. Executive Summary

The Office of the Public Service Commission (OPSC) of Vanuatu is responsible for the oversight and regulation of public service employment matters, including recruitment, promotions, transfers, discipline, and organisational structure changes. Currently, the processing of ministry submissions through commission deliberation, decision recording, and implementation tracking is managed through paper-based workflows, resulting in processing delays, limited visibility, and administrative overhead.

The Commission Decision Management System (SCDMS) has been developed to digitise and automate the end-to-end submission lifecycle. The system provides a secure, role-based digital platform that enables ministry users to submit applications electronically, PSC officers to assess and process them through defined workflow stages, commissioners to deliberate and record decisions during sittings, and post-decision officers to track implementation to completion.

Built on modern open-source technologies and containerised for efficient deployment, SCDMS delivers a fully auditable, NCSS 2030-compliant solution that reduces processing times, enhances transparency, and strengthens accountability across the entire PSC decision-making value chain.

2. Project Background & Objectives

2.1 Current State

The PSC currently processes an average of 80–120 submissions per quarter across 15 ministries. The existing paper-based process presents several challenges:

- **Limited visibility:** Manual tracking of submission status across multiple units
- **Fragmented records:** No central repository for decisions, minutes, or implementation records
- **Delayed processing:** Difficulties in monitoring assessment timelines and statutory deadlines
- **Communication gaps:** No automated notifications or escalation mechanisms
- **Reporting burden:** Laborious generation of reports and performance metrics

2.2 Objectives

- Digitise the complete submission lifecycle from intake through to implementation reporting
- Enforce standardised workflow stages with role-based transition controls
- Provide real-time submission tracking and status visibility to all stakeholders
- Enable electronic commission deliberation and decision recording
- Implement comprehensive audit trails for accountability and transparency
- Facilitate data-driven reporting on processing times, bottlenecks, and outcomes

3. Scope of Work

3.1 In Scope

The system covers the following functional areas:

Module	Functions	Primary Users
Submission Management	Drafting, submission, receipt, registration, routing	Ministry HR, PSC Officer
Assessment Pipeline	Checklist review, 21-day assessment, clarification loops, deferrals	Unit Manager, PSC Officer
Commission Deliberation	Agenda management, meeting scheduling, decision recording	Secretary, Commissioner
Post-Decision	Minutes drafting, formal notification, task allocation	Secretary, OPSC Manager
Implementation Tracking	Task assignment, status updates, close-out reporting	Principal / Senior Officer
Administration	User management, role-based permissions, system configuration	PSC Admin
Reporting & Analytics	Dashboards, export, processing time metrics, audit logs	All users

3.2 Out of Scope

- Payroll or human resource information system integration
- Document scanning or optical character recognition
- Mobile-native applications (the system is web-responsive)

4. System Architecture

SCDMS follows a modern, containerised microservices architecture:

Component	Technology	Purpose	Container
Backend API	Django 6 / Python 3.12	Business logic, workflow engine, REST API	Yes
Web Server	nginx 1.27	Static files, API reverse proxy, TLS termination	Yes
Frontend	React 18 / Vite / Tailwind CSS	Web-based user interface	Yes
Database	PostgreSQL 16	All structured application data	Yes
Message Broker	Redis 7	Async task queue for background jobs	Yes
Background Workers	Celery	Email notifications, scheduled backups, AI feedback	Yes

4.1 Workflow Engine

The core of the system is a configurable workflow engine that enforces stage-to-stage transitions based on user role and submission state. The engine is defined as a directed graph with 20+ stages and supports conditional branching, clarification loops, parallel legal review paths, and automated deadline calculations (e.g., 21 working days for assessment).

5. User Profile & Access Model

User Group	Role	Count	Access
PSC Administrator	psc_admin	1–2	Full system access
PSC Officer	psc_officer	3–5	Intake, assessment, workflow transitions
PSC Secretary	psc_secretary	1–2	Meeting management, agenda, minutes
Commissioner	psc_commissioner	5–7	Deliberation, decision recording
OPSC Manager	psc_manager	2–3	Post-decision task allocation
Principal / Senior Officer	principal_officer / senior_officer	3–5	Implementation execution
VIPAM / HR / Compliance Manager	vipam_manager / hr_unit_manager / compliance_manager	3–5	Checklist review per routed unit
Ministry HR Officer	ministry_hr	15	Submission drafting and submission
Department Admin	dept_admin	Variable	View-only departmental access

Access to each workflow transition is governed by a role-based permission matrix defined at the code level and configurable through the administration panel.

6. Security Framework

The system has been designed in accordance with the Vanuatu NCSS 2030 cybersecurity standards. The following controls are implemented:

- **Authentication:** JWT token-based authentication with automatic expiry and refresh rotation. Two-factor authentication (TOTP) enforced globally with individual user opt-out available.
- **Access Control:** 12 distinct roles with granular permission sets. Every API endpoint and workflow transition is validated against the requesting user's role.
- **Account Protection:** Django-axes account lockout after configurable failed attempts. Login rate-limited to 5 requests per minute per IP address.
- **Audit Trail:** All stage transitions are logged immutably with actor identity, timestamp, previous and new stage, and optional remarks. Admin actions are separately logged in a dedicated audit log.
- **Data Protection:** Database password and secret key stored in environment variables. HTTPS-ready architecture. Security headers (CSP, HSTS, X-Frame-Options, X-Content-Type-Options) enforced at the nginx reverse proxy layer.
- **Password Policy:** Configurable minimum length, character complexity requirements, and password history enforcement to prevent reuse.

7. Deployment Strategy

7.1 Infrastructure Requirements

Resource	Requirement
Server Type	Virtual Machine
Operating System	Linux (Debian/Ubuntu recommended)
vCPU	4 cores
Memory	16 GB RAM
Disk (OS)	100 GB SSD
Data Volume	500 GB SSD
Docker Engine	Version 24+
Network	Public IP with DNS + HTTPS certificate

7.2 Deployment Model

The application is fully containerised using Docker Compose, enabling consistent deployment across development, staging, and production environments. A single "docker-compose up" command initialises all six services with no manual configuration beyond the .env file.

7.3 Backup & Recovery

Automated database backups can be scheduled via the administration panel with configurable retention periods. Backup files are stored on the data volume and can be downloaded for off-site storage. A restore function supports both file-based and upload-based recovery.

8. Benefits & Outcomes

- **Process Efficiency:** Eliminates paper-based tracking; automated routing reduces processing time from weeks to days for standard submissions.
- **Transparency & Visibility:** Ministries can track their submissions in real time. All stakeholders have visibility into current status, assessment progress, and decision outcomes.
- **Accountability:** Every action is logged with actor identity and timestamp. Role-based access ensures that only authorised users can perform stage transitions.
- **Data-Driven Oversight:** Built-in analytics provide insight into processing times, bottlenecks, unit workloads, and decision trends — enabling evidence-based process improvements.
- **Audit Readiness:** Complete electronic audit trail satisfies internal and external audit requirements. No more manual compilation of paper records.
- **Cost Savings:** Reduced administrative overhead, elimination of physical document transport, and faster decision cycles deliver measurable operational savings.